



e Impact Influence

Configuration des équipements

1. Plan d'adressage IP :

Adresse IPv4	132.186.32.32/27	255.255.255.224
Plage d'adresses		

Adresse réseau	IP Début	IP Fin	Nombre adresses IP disponibles	Adresse de diffusion	Masque réseau
132.186.32.32	132.186.32.33	132.186.32.34	2	132.186.32.35	/30
132.186.32.36	132.186.32.37	132.186.32.38	2	132.186.32.39	/30
132.186.32.40	132.168.32.41	132.168.32.42	2	132.186.32.43	/30
132.186.32.44	132.186.32.45	132.186.32.46	2	132.186.32.47	/30

N°VLAN	VLAN	Adresse réseau	IPv4 Début	IPv4 Fin	IP disponibles	Adresse de diffusion
20	INFORMATIQUE	192.168.20.0/24	192.168.20.1	192.168.20.254	254	192.168.20.255
21	FINANCE	192.168.21.0/24	192.168.21.1	192.168.21.254	254	192.168.21.255
22	COMMERCIAL	192.168.22.0/24	192.168.22.1	192.168.22.254	254	192.168.22.255
23	RH	192.168.23.0/24	192.168.23.1	192.168.23.254	254	192.168.23.255
24	DIRECTION	192.168.24.0/24	192.168.24.1	192.168.24.254	254	192.168.24.255
25	SERVEUR DNS	192.168.25.0/24	192.168.25.1	192.168.25.254	254	192.168.25.255
	LAN					
	SERVEUR WEB	192.168.30.0/24	192.168.30.1	192.168.30.254	254	192.168.30.255
	REUNION	192.168.31.0/24	192.168.31.1	192.168.31.254	254	192.168.31.255
	ACCUEIL	192.168.40.0/24	192.168.40.1	192.168.40.254	254	192.168.40.255
	SHOWROOM	192.168.41.0/24	192.168.41.1	192.168.41.254	254	192.168.41.255

Équipement	Interface	Adresse IPv4	Adresse IPv6	IPv6 Link-local	Commentaire
Routeur Rouge	Serial0/0/0	132.186.32.34/30	2001:0:CAFE:EEEE::1/66	FE80:EEEE::1	Connexion vers Routeur Bleu
Routeur Rouge	Serial0/0/1	132.186.32.41/30	2001:0:CAFE:BEEE::1/66	FE80:BEEE::1	vers Routeur Vert
Routeur Rouge	Gig0/0.20	192.168.20.254	2001:0:CAFE:BE20::254	FE80:0:CAFE:BE20::1	Vlan 20 Informatique
Routeur Rouge	Gig0/0.21	192.168.21.254	2001:0:CAFE:BE21::254	FE80:0:CAFE:BE21::1	Vlan 21 Finance
Routeur Rouge	Gig0/0.22	192.168.22.254	2001:0:CAFE:BE22::254	FE80:0:CAFE:BE22::1	Vlan 22 Commercial
Routeur Rouge	Gig0/0.23	192.168.23.254	2001:0:CAFE:BE23::254	FE80:0:CAFE:BE23::1	Vlan 23 RH
Routeur Rouge	Gig0/0.24	192.168.24.254	2001:0:CAFE:BE24::254	FE80:0:CAFE:BE24::1	Vlan 24 Direction
Routeur Rouge	Gig0/0.25	192.168.25.254	2001:0:CAFE:BE25:254	FE80:0:CAFE:BE25::1	Vlan 25 Serveur-Rouge
L3 Switch Core 00	Vlan 20	192.168.20.253	2001:0:CAFE:BE20::253		Vlan Informatique
L3 Switch Core 00	Vlan 21	192.168.21.253	2001:0:CAFE:BE21::253		Vlan Finance
L3 Switch Core 00	Vlan 22	192.168.22.253	2001:0:CAFE:BE22::253		Vlan Commercial
L3 Switch Core 00	Vlan 23	192.168.23.253	2001:0:CAFE:BE23::253		Vlan RH
L3 Switch Core 00	Vlan 24	192.168.24.253	2001:0:CAFE:BE24::253		Vlan Direction

L3 Switch Core 00	Vlan 25	192.168.25.253	2001:0:CAFE:BE25::253		Vlan Serveur Bat Rouge
L3 Switch Distributed 00	Vlan 20	192.168.20.252	2001:0:CAFE:BE20::252		Vlan Informatique
L3 Switch Distributed 00	Vlan 21	192.168.21.252	2001:0:CAFE:BE21::252		Vlan Finance
L3 Switch Distributed 00	Vlan 22	192.168.22.252	2001:0:CAFE:BE22::252		Vlan Commercial
L3 Switch Distributed 00	Vlan 23	192.168.23.252	2001:0:CAFE:BE23::252		Vlan RH
L3 Switch Distributed 00	Vlan 24	192.168.24.252	2001:0:CAFE:BE24::252		Vlan Direction
L3 Switch Distributed 00	Vlan 25	192.168.25.252	2001:0:CAFE:BE25::252		Vlan Serveur Bat Rouge
L3 Switch Distributed 01	Vlan 20	192.168.20.251	2001:0:CAFE:BE20::251		Vlan Informatique
L3 Switch Distributed 01	Vlan 21	192.168.21.251	2001:0:CAFE:BE21::251		Vlan Finance
L3 Switch Distributed 01	Vlan 22	192.168.22.251	2001:0:CAFE:BE22::251		Vlan Commercial
L3 Switch Distributed 01	Vlan 23	192.168.23.251	2001:0:CAFE:BE23::251		Vlan RH
L3 Switch Distributed 01	Vlan 24	192.168.24.251	2001:0:CAFE:BE24::251		Vlan Direction
L3 Switch Distributed 01	Vlan 25	192.168.25.251	2001:0:CAFE:BE25::251		Vlan Serveur Bat Rouge
Routeur Vert	Serial0/0/0	132.186.32.37/30	2001:0:CAFE:EDEE::1/66	FE80:EDEE::1	vers Routeur Bleu
Routeur Vert	Serial0/0/1	132.186.32.42/30	2001:0:CAFE:BEEE::2/66	FE80:BEEE::2	vers Routeur Rouge
Routeur Vert	Gig0/0	192.168.30.254	2001:0:CAFE:ED30::254	FE80:0:CAFE:ED30::1	vers Switch Bat Vert Serveur
Routeur Vert	Gig0/1	192.168.31.254	2001:0:CAFE:ED31::254		vers Switch Reunion
Routeur Bleu	Serial0/0/0	132.186.32.33	2001:0:CAFE:EEEE::2/66	FE80:EEEE::2	vers Routeur Rouge
Routeur Bleu	Serial0/0/1	132.186.32.38	2001:0:CAFE:EDEE::2/66	FE80:EDEE::2	vers Routeur Vert
Routeur Bleu	Gig0/0	192.168.40.254	2001:0:CAFE:EE40::254		vers Switch Accueil
Routeur Bleu	Gig0/1	192.168.41.254	2001:0:CAFE:EE41::254		vers Switch Showroom
Serveur WEB	Fa0	192.168.30.2	2001:0:CAFE:ED30::2		vers Switch Bat Vert Serveur
DNS Serveur	Fa0	192.168.25.2	2001:0:CAFE:BE25::2		vers Switch Bat Rouge Serveur

2. Configuration DNS :

Server DNS

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type **A Record** ▼

Address

No.	Name	Type	Detail
0	impactinfluence.com	A Record	132.186.32.37
1	impactinfluencev6.com	AAAA Record	2001:0:CAFE:ED30::2

3. Configuration des routeurs (comprenant les tables de routage, les sous-interfaces et les NAT) :

Bâtiment Bleu :

- hostname Router-bat-bleu
- ip dhcp pool ACCUEIL
 - network 192.168.40.0 255.255.255.0
 - default-router 192.168.40.254
 - dns-server 192.168.25.2
- ip dhcp pool SHOWROOM
 - network 192.168.41.0 255.255.255.0
 - default-router 192.168.41.254
 - dns-server 192.168.25.2
- ipv6 unicast-routing
- interface GigabitEthernet0/0
 - description "vers Switch Accueil"
 - ip address 192.168.40.254 255.255.255.0
 - ip nat inside
 - duplex auto
 - speed auto
 - ipv6 address 2001:0:CAFE:EE40::254/64
 - ipv6 eigrp 1

```

- interface GigabitEthernet0/1
    description "vers Switch Showroom"
    ip address 192.168.41.254 255.255.255.0
    ip nat inside
    duplex auto
    speed auto
    ipv6 address 2001:0:CAFE:EE41::254/64
    ipv6 eigrp 1

- interface Serial0/0/0
    description "vers Routeur Bat Rouge"
    ip address 132.186.32.33 255.255.255.252
    ip nat outside
    ipv6 address FE80:EEEE::2 link-local
    ipv6 address 2001:0:CAFE:EEEE::2/66
    ipv6 eigrp 1
    clock rate 2000000

- interface Serial0/0/1
    description "vers Routeur Bat Vert"
    ip address 132.186.32.38 255.255.255.252
    ip nat outside
    ipv6 address FE80:EDEE::2 link-local
    ipv6 address 2001:0:CAFE:EDEE::2/66
    ipv6 eigrp 1
    clock rate 2000000

- router eigrp 1
    network 132.186.32.36 0.0.0.3
    network 132.186.32.32 0.0.0.3

- ipv6 router eigrp 1
    eigrp router-id 3.3.3.3
    no shutdown
    passive-interface GigabitEthernet0/0
    passive-interface GigabitEthernet0/1

- ip nat inside source list 40 interface Serial0/0/1 overload
- ip nat inside source list 41 interface Serial0/0/1 overload

- access-list 40 permit 192.168.40.0 0.0.0.255
- access-list 41 permit 192.168.41.0 0.0.0.255

- show ip route :
    Gateway of last resort is not set

```

132.186.0.0/16 is variably subnetted, 5 subnets, 2 masks
 C 132.186.32.32/30 is directly connected, Serial0/0/0
 L 132.186.32.33/32 is directly connected, Serial0/0/0
 C 132.186.32.36/30 is directly connected, Serial0/0/1
 L 132.186.32.38/32 is directly connected, Serial0/0/1
 D 132.186.32.40/30 [90/2681856] via 132.186.32.37, 00:12:47, Serial0/0/1
 [90/2681856] via 132.186.32.34, 00:12:46, Serial0/0/0
 192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
 C 192.168.40.0/24 is directly connected, GigabitEthernet0/0
 L 192.168.40.254/32 is directly connected, GigabitEthernet0/0
 192.168.41.0/24 is variably subnetted, 2 subnets, 2 masks
 C 192.168.41.0/24 is directly connected, GigabitEthernet0/1
 L 192.168.41.254/32 is directly connected, GigabitEthernet0/1

- show ipv6 route :

```

D          2001:0:CAFE:BEEE::/66 [90/2681856]
            via FE80:EDEE::1, Serial0/0/1
            via FE80:EEEE::1, Serial0/0/0
D          2001:0:CAFE:ED30::/64 [90/2170112]
            via FE80:EDEE::1, Serial0/0/1
D          2001:0:CAFE:ED31::/64 [90/2170112]
            via FE80:EDEE::1, Serial0/0/1
C          2001:0:CAFE:EDEE::/66 [0/0]
            via Serial0/0/1, directly connected
L          2001:0:CAFE:EDEE::2/128 [0/0]
            via Serial0/0/1, receive
C          2001:0:CAFE:EE40::/64 [0/0]
            via GigabitEthernet0/0, directly connected
L          2001:0:CAFE:EE40::254/128 [0/0]
            via GigabitEthernet0/0, receive
C          2001:0:CAFE:EE41::/64 [0/0]
            via GigabitEthernet0/1, directly connected
L          2001:0:CAFE:EE41::254/128 [0/0]
            via GigabitEthernet0/1, receive
C          2001:0:CAFE:EEEE::/66 [0/0]
            via Serial0/0/0, directly connected
L          2001:0:CAFE:EEEE::2/128 [0/0]
            via Serial0/0/0, receive
L          FF00::/8 [0/0]
            via Null0, receive
  
```

Bâtiment Rouge :

- hostname Router-bat-rouge

- ipv6 unicast-routing

- ipv6 dhcp pool VLAN20
 - dns-server 2001:0:CAFE:BE25::2

- ipv6 dhcp pool VLAN21
 - dns-server 2001:0:CAFE:BE25::2

- ipv6 dhcp pool VLAN22
 - dns-server 2001:0:CAFE:BE25::2

- ipv6 dhcp pool VLAN23
 - dns-server 2001:0:CAFE:BE25::2

- ipv6 dhcp pool VLAN24
 - dns-server 2001:0:CAFE:BE25::2

- interface GigabitEthernet0/0.20
 - description "vlan 20 Informatique"
 - encapsulation dot1Q 20
 - ip address 192.168.20.254 255.255.255.0
 - ip access-group VlanIT in
 - ip nat inside
 - ipv6 address FE80:0:CAFE:BE20::1 link-local
 - ipv6 address 2001:0:CAFE:BE20::254/64
 - ipv6 nd other-config-flag
 - ipv6 dhcp server VLAN20

- interface GigabitEthernet0/0.21
 - description "vlan 21 Finance"
 - encapsulation dot1Q 21
 - ip address 192.168.21.254 255.255.255.0
 - ip access-group VlanFinance in
 - ip nat inside
 - ipv6 address FE80:0:CAFE:BE21::1 link-local
 - ipv6 address 2001:0:CAFE:BE21::254/64
 - ipv6 nd other-config-flag
 - ipv6 dhcp server VLAN21

- interface GigabitEthernet0/0.22
 - description "vlan 22 Commercial"
 - encapsulation dot1Q 22
 - ip address 192.168.22.254 255.255.255.0
 - ip access-group VlanCommercial in
 - ip nat inside
 - ipv6 address FE80:0:CAFE:BE22::1 link-local

```
ipv6 address 2001:0:CAFE:BE22::254/64
ipv6 nd other-config-flag
ipv6 dhcp server VLAN22
```

```
- interface GigabitEthernet0/0.23
  description "vlan 23 RH"
  encapsulation dot1Q 23
  ip address 192.168.23.254 255.255.255.0
  ip access-group VlanRH in
  ip nat inside
  ipv6 address FE80:0:CAFE:BE23::1 link-local
  ipv6 address 2001:0:CAFE:BE23::254/64
  ipv6 nd other-config-flag
  ipv6 dhcp server VLAN23
```

```
- interface GigabitEthernet0/0.24
  description "vlan 24 Direction"
  encapsulation dot1Q 24
  ip address 192.168.24.254 255.255.255.0
  ip access-group VlanDirection in
  ip nat inside
  ipv6 address FE80:0:CAFE:BE24::1 link-local
  ipv6 address 2001:0:CAFE:BE24::254/64
  ipv6 nd other-config-flag
  ipv6 dhcp server VLAN24
```

```
- interface GigabitEthernet0/0.25
  description "vlan 25 Serveur Bat Rouge"
  encapsulation dot1Q 25
  ip address 192.168.25.254 255.255.255.0
  ip access-group VlanDNS in
  ipv6 address FE80:0:CAFE:BE25::1 link-local
  ipv6 address 2001:0:CAFE:BE25::254/64
```

```
- interface Serial0/0/0
  description "vers Routeur Bat Bleu"
  ip address 132.186.32.34 255.255.255.252
  ip nat outside
  ipv6 address FE80:EEEE::1 link-local
  ipv6 address 2001:0:CAFE:EEEE::1/66
  ipv6 eigrp 1
```

```
- interface Serial0/0/1
  description "vers Routeur Bat Vert"
  ip address 132.186.32.41 255.255.255.252
```

```
ip nat outside
ipv6 address FE80:BEEE::1 link-local
ipv6 address 2001:0:CAFE:BEEE::1/66
ipv6 eigrp 1
```

```
- router eigrp 1
  network 132.186.32.40 0.0.0.3
  network 132.186.32.32 0.0.0.3

- ipv6 router eigrp 1
  eigrp router-id 1.1.1.1
  no shutdown
  passive-interface GigabitEthernet0/0
```

```
- ip nat inside source list VLAN20 interface Serial0/0/1 overload
- ip nat inside source list VLAN21 interface Serial0/0/1 overload
- ip nat inside source list VLAN22 interface Serial0/0/1 overload
- ip nat inside source list VLAN23 interface Serial0/0/1 overload
- ip nat inside source list VLAN24 interface Serial0/0/1 overload
- ip nat inside source list VLAN25 interface Serial0/0/1 overload
```

```
- ip access-list standard VLAN20
  permit 192.168.20.0 0.0.0.255
- ip access-list standard VLAN21
  permit 192.168.21.0 0.0.0.255
- ip access-list standard VLAN22
  permit 192.168.22.0 0.0.0.255
- ip access-list standard VLAN23
  permit 192.168.23.0 0.0.0.255
- ip access-list standard VLAN24
  permit 192.168.24.0 0.0.0.255
- ip access-list standard VLAN25
  permit 192.168.25.0 0.0.0.255
```

```
- ip access-list extended VlanIT
  permit udp 192.168.20.0 0.0.0.255 host 192.168.25.2 eq domain
  permit tcp 192.168.20.0 0.0.0.255 host 132.186.32.37 eq www
  permit icmp 192.168.20.0 0.0.0.255 192.168.21.0 0.0.0.255 echo
  permit icmp 192.168.20.0 0.0.0.255 192.168.22.0 0.0.0.255 echo
  permit icmp 192.168.20.0 0.0.0.255 192.168.23.0 0.0.0.255 echo
  permit icmp 192.168.20.0 0.0.0.255 192.168.24.0 0.0.0.255 echo
  permit icmp 192.168.20.0 0.0.0.255 192.168.25.0 0.0.0.255 echo
  permit icmp 192.168.20.0 0.0.0.255 host 132.186.32.37
  permit icmp 192.168.20.0 0.0.0.255 host 132.186.32.38
  permit icmp 192.168.20.0 0.0.0.255 host 132.186.32.41
```

- ip access-list extended VlanFinance

```
permit udp 192.168.21.0 0.0.0.255 host 192.168.25.2 eq domain
permit tcp 192.168.21.0 0.0.0.255 host 192.168.30.2 eq www
permit tcp 192.168.21.0 0.0.0.255 host 132.186.32.37 eq www
permit icmp 192.168.21.0 0.0.0.255 192.168.20.0 0.0.0.255 echo-reply
permit icmp 192.168.21.0 0.0.0.255 192.168.22.0 0.0.0.255
permit icmp 192.168.21.0 0.0.0.255 192.168.23.0 0.0.0.255
permit icmp 192.168.21.0 0.0.0.255 192.168.24.0 0.0.0.255
permit icmp 192.168.21.0 0.0.0.255 host 132.186.32.37
permit icmp 192.168.21.0 0.0.0.255 host 132.186.32.38
permit icmp 192.168.21.0 0.0.0.255 host 132.186.32.41
```

- ip access-list extended VlanCommercial

```
permit udp 192.168.22.0 0.0.0.255 host 192.168.25.2 eq domain
permit tcp 192.168.22.0 0.0.0.255 host 192.168.30.2 eq www
permit tcp 192.168.22.0 0.0.0.255 host 132.186.32.37 eq www
permit icmp 192.168.22.0 0.0.0.255 192.168.20.0 0.0.0.255 echo-reply
permit icmp 192.168.22.0 0.0.0.255 192.168.21.0 0.0.0.255
permit icmp 192.168.22.0 0.0.0.255 192.168.23.0 0.0.0.255
permit icmp 192.168.22.0 0.0.0.255 192.168.24.0 0.0.0.255
permit icmp 192.168.22.0 0.0.0.255 host 132.186.32.37
permit icmp 192.168.22.0 0.0.0.255 host 132.186.32.38
permit icmp 192.168.22.0 0.0.0.255 host 132.186.32.41
```

- ip access-list extended VlanRH

```
permit udp 192.168.23.0 0.0.0.255 host 192.168.25.2 eq domain
permit tcp 192.168.23.0 0.0.0.255 host 192.168.30.2 eq www
permit tcp 192.168.23.0 0.0.0.255 host 132.186.32.37 eq www
permit icmp 192.168.23.0 0.0.0.255 192.168.20.0 0.0.0.255 echo-reply
permit icmp 192.168.23.0 0.0.0.255 192.168.21.0 0.0.0.255
permit icmp 192.168.23.0 0.0.0.255 192.168.22.0 0.0.0.255
permit icmp 192.168.23.0 0.0.0.255 192.168.24.0 0.0.0.255
permit icmp 192.168.23.0 0.0.0.255 host 132.186.32.37
permit icmp 192.168.23.0 0.0.0.255 host 132.186.32.38
permit icmp 192.168.23.0 0.0.0.255 host 132.186.32.41
```

- ip access-list extended VlanDirection

```
permit udp 192.168.24.0 0.0.0.255 host 192.168.25.2 eq domain
permit tcp 192.168.24.0 0.0.0.255 host 192.168.30.2 eq www
permit tcp 192.168.24.0 0.0.0.255 host 132.186.32.37 eq www
permit icmp 192.168.24.0 0.0.0.255 192.168.20.0 0.0.0.255 echo-reply
permit icmp 192.168.24.0 0.0.0.255 192.168.21.0 0.0.0.255
permit icmp 192.168.24.0 0.0.0.255 192.168.22.0 0.0.0.255
permit icmp 192.168.24.0 0.0.0.255 192.168.23.0 0.0.0.255
permit icmp 192.168.24.0 0.0.0.255 host 132.186.32.37
permit icmp 192.168.24.0 0.0.0.255 host 132.186.32.38
permit icmp 192.168.24.0 0.0.0.255 host 132.186.32.41
```

- ip access-list extended VlanDNS


```

permit icmp 192.168.25.0 0.0.0.255 192.168.20.0 0.0.0.255 echo-reply
permit udp host 192.168.25.2 eq domain 192.168.20.0 0.0.0.255
permit udp host 192.168.25.2 eq domain 192.168.21.0 0.0.0.255
permit udp host 192.168.25.2 eq domain 192.168.22.0 0.0.0.255
permit udp host 192.168.25.2 eq domain 192.168.23.0 0.0.0.255
permit udp host 192.168.25.2 eq domain 192.168.24.0 0.0.0.255

```

- sh ip route :

Gateway of last resort is not set

```

132.186.0.0/16 is variably subnetted, 5 subnets, 2 masks
C       132.186.32.32/30 is directly connected, Serial0/0/0
L       132.186.32.34/32 is directly connected, Serial0/0/0
D       132.186.32.36/30 [90/2681856] via 132.186.32.42, 00:01:55, Serial0/0/1
        [90/2681856] via 132.186.32.33, 00:01:55, Serial0/0/0
C       132.186.32.40/30 is directly connected, Serial0/0/1
L       132.186.32.41/32 is directly connected, Serial0/0/1
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.254/32 is directly connected, GigabitEthernet0/0.20
192.168.21.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.21.0/24 is directly connected, GigabitEthernet0/0.21
L       192.168.21.254/32 is directly connected, GigabitEthernet0/0.21
192.168.22.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.22.0/24 is directly connected, GigabitEthernet0/0.22
L       192.168.22.254/32 is directly connected, GigabitEthernet0/0.22
192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.23.0/24 is directly connected, GigabitEthernet0/0.23
L       192.168.23.254/32 is directly connected, GigabitEthernet0/0.23
192.168.24.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.24.0/24 is directly connected, GigabitEthernet0/0.24
L       192.168.24.254/32 is directly connected, GigabitEthernet0/0.24
192.168.25.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.25.0/24 is directly connected, GigabitEthernet0/0.25
L       192.168.25.254/32 is directly connected, GigabitEthernet0/0.25

```

- sh ipv6 route :

```

C       2001:0:CAFE:BE20::/64 [0/0]
        via GigabitEthernet0/0.20, directly connected
L       2001:0:CAFE:BE20::254/128 [0/0]
        via GigabitEthernet0/0.20, receive
C       2001:0:CAFE:BE21::/64 [0/0]
        via GigabitEthernet0/0.21, directly connected
L       2001:0:CAFE:BE21::254/128 [0/0]
        via GigabitEthernet0/0.21, receive
C       2001:0:CAFE:BE22::/64 [0/0]

```

```

                via GigabitEthernet0/0.22, directly connected
L      2001:0:CAFE:BE22::254/128 [0/0]
                via GigabitEthernet0/0.22, receive
C      2001:0:CAFE:BE23::/64 [0/0]
                via GigabitEthernet0/0.23, directly connected
L      2001:0:CAFE:BE23::254/128 [0/0]
                via GigabitEthernet0/0.23, receive
C      2001:0:CAFE:BE24::/64 [0/0]
                via GigabitEthernet0/0.24, directly connected
L      2001:0:CAFE:BE24::254/128 [0/0]
                via GigabitEthernet0/0.24, receive
C      2001:0:CAFE:BE25::/64 [0/0]
                via GigabitEthernet0/0.25, directly connected
L      2001:0:CAFE:BE25::254/128 [0/0]
                via GigabitEthernet0/0.25, receive
C      2001:0:CAFE:BEEE::/66 [0/0]
                via Serial0/0/1, directly connected
L      2001:0:CAFE:BEEE::1/128 [0/0]
                via Serial0/0/1, receive
D      2001:0:CAFE:ED30::/64 [90/2170112]
                via FE80:BEEE::2, Serial0/0/1
D      2001:0:CAFE:ED31::/64 [90/2170112]
                via FE80:BEEE::2, Serial0/0/1
D      2001:0:CAFE:EDEE::/66 [90/2681856]
                via FE80:EEEE::2, Serial0/0/0
                via FE80:BEEE::2, Serial0/0/1
D      2001:0:CAFE:EE40::/64 [90/2170112]
                via FE80:EEEE::2, Serial0/0/0
D      2001:0:CAFE:EE41::/64 [90/2170112]
                via FE80:EEEE::2, Serial0/0/0
C      2001:0:CAFE:EEEE::/66 [0/0]
                via Serial0/0/0, directly connected
L      2001:0:CAFE:EEEE::1/128 [0/0]
                via Serial0/0/0, receive
L      FF00::/8 [0/0]
                via Null0, receive

```

Bâtiment Vert :

```

- hostname Router-bat-vert
- ip dhcp pool REUNION
    network 192.168.31.0 255.255.255.0
    default-router 192.168.31.254
    dns-server 192.168.25.2

```

- ipv6 unicast-routing
- interface GigabitEthernet0/0
 - description "vers Switch Serveur Bat Rouge"
 - ip address 192.168.30.254 255.255.255.0
 - ip nat inside
 - duplex auto
 - speed auto
 - ipv6 address FE80:0:CAFE:ED30::1 link-local
 - ipv6 address 2001:0:CAFE:ED30::254/64
 - ipv6 eigrp 1
- interface GigabitEthernet0/1
 - description "vers Switch Reunion"
 - ip address 192.168.31.254 255.255.255.0
 - duplex auto
 - speed auto
 - ipv6 address 2001:0:CAFE:ED31::254/64
 - ipv6 eigrp 1
- interface Serial0/0/0
 - description "vers Routeur Batiment Bleu"
 - ip address 132.186.32.37 255.255.255.252
 - ip nat outside
 - ipv6 address FE80:EDEE::1 link-local
 - ipv6 address 2001:0:CAFE:EDEE::1/66
 - ipv6 eigrp 1
- interface Serial0/0/1
 - description "vers Routeur Batiment Rouge"
 - ip address 132.186.32.42 255.255.255.252
 - ip nat outside
 - ipv6 address FE80:BEEE::2 link-local
 - ipv6 address 2001:0:CAFE:BEEE::2/66
 - ipv6 eigrp 1
 - clock rate 2000000
- router eigrp 1
 - network 132.186.32.36 0.0.0.3
 - network 132.186.32.40 0.0.0.3
- ipv6 router eigrp 1
 - eigrp router-id 2.2.2.2
 - no shutdown
 - passive-interface GigabitEthernet0/0

```
passive-interface GigabitEthernet0/1
```

- ip nat inside source list WEB interface Serial0/0/0 overload
- ip nat inside source static tcp 192.168.30.2 80 132.186.32.37 80
- ip nat inside source static 192.168.30.2 132.186.32.37

- ```
- ip access-list standard WEB
 permit 192.168.30.0 0.0.0.255
```

- show ip route :

Gateway of last resort is not set

132.186.0.0/16 is variably subnetted, 5 subnets, 2 masks

D 132.186.32.32/30 [90/2681856] via 132.186.32.41, 00:03:09, Serial0/0/1  
[90/2681856] via 132.186.32.38, 00:03:08, Serial0/0/0

```
C 132.186.32.36/30 is directly connected, Serial0/0/0
```

```
L 132.186.32.37/32 is directly connected, Serial0/0/0
```

```
C 132.186.32.40/30 is directly connected, Serial0/0/1
```

```
L 132.186.32.42/32 is directly connected, Serial0/0/1
```

192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks

```
C 192.168.30.0/24 is directly connected, GigabitEthernet0/0
```

```
L 192.168.30.254/32 is directly connected, GigabitEthernet0/0
```

192.168.31.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.31.0/24 is directly connected, GigabitEthernet0/1

```
L 192.168.31.254/32 is directly connected, GigabitEthernet0/1
```

- show ipv6 route :

```
C 2001:0:CAFE:BEEE::66 [0/0]
 via Serial0/0/1, directly connected
```

```
L 2001:0:CAFE:BEEE::2/128 [0/0]
 via Serial0/0/1, receive
```

```
C 2001:0:CAFE:ED30::/64 [0/0]
 via GigabitEthernet0/0, directly connected
```

```
L 2001:0:CAFE:ED30::254/128 [0/0]
 via GigabitEthernet0/0, receive
```

```
C 2001:0:CAFE:ED31::/64 [0/0]
 via GigabitEthernet0/1, directly connected
```

```
L 2001:0:CAFE:ED31::254/128 [0/0]
 via GigabitEthernet0/1, receive
```

```
C 2001:0:CAFE:EDEE::/66 [0/0]
 via Serial0/0/0, directly connected
```

```
L 2001:0:CAFE:EDEE::1/128 [0/0]
 via Serial0/0/0, receive
```

```
D 2001:0:CAFE:EE40::/64 [90/2170112]
 via FE80:EDEE::2, Serial0/0/0
```

D 2001:0:CAFE:EE41::/64 [90/2170112]

```

 via FE80:EDEE::2, Serial0/0/0
D 2001:0:CAFE:EEEE::/66 [90/2681856]
 via FE80:BEEE::1, Serial0/0/1
 via FE80:EDEE::2, Serial0/0/0
L FF00::/8 [0/0]
 via Null0, receive

```

#### **4. Configuration des switchs (comprenant les paramètres DHCP, les VLAN, le LACP et les ACL) :**

##### **4.1 Configuration des switchs de type layer 3 :**

###### **Switch Core :**

- hostname sw-core-00
- ip routing
- ipv6 unicast-routing
- interface GigabitEthernet1/0/1
  - description "vers Routeur Bat Rouge"
  - switchport trunk allowed vlan 20-25
  - switchport mode trunk
- interface GigabitEthernet1/0/2
  - description "vers Switch Distributed 00"
  - switchport trunk allowed vlan 20-25
  - switchport mode trunk
- interface GigabitEthernet1/0/3
  - description "vers Switch Distributed 01"
  - switchport trunk allowed vlan 20-25
  - switchport mode trunk
- interface Vlan20
  - description "vlan Informatique"
  - mac-address 0090.2b42.4801
  - ip address 192.168.20.253 255.255.255.0
  - ipv6 address 2001:0:CAFE:BE20::253/64
- interface Vlan21
  - description "vlan Finance"
  - mac-address 0090.2b42.4802

```
ip address 192.168.21.253 255.255.255.0
ipv6 address 2001:0:CAFE:BE21::253/64
```

```
- interface Vlan22
 description "vlan Commercial"
 mac-address 0090.2b42.4803
 ip address 192.168.22.253 255.255.255.0
 ipv6 address 2001:0:CAFE:BE22::253/64
```

```
- interface Vlan23
 description "vlan RH"
 mac-address 0090.2b42.4804
 ip address 192.168.23.253 255.255.255.0
 ipv6 address 2001:0:CAFE:BE23::253/64
```

```
- interface Vlan24
 description "vlan Direction"
 mac-address 0090.2b42.4805
 ip address 192.168.24.253 255.255.255.0
 ipv6 address 2001:0:CAFE:BE24::253/64
```

```
- interface Vlan25
 description "vlan Serveur Bat Rouge"
 mac-address 0090.2b42.4806
 ip address 192.168.25.253 255.255.255.0
 ipv6 address 2001:0:CAFE:BE25::253/64
```

### **Switch Distributed 1 :**

```
- hostname sw-distri-00

- ip dhcp pool VLAN20
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.254
 dns-server 192.168.25.2

- ip dhcp pool VLAN21
 network 192.168.21.0 255.255.255.0
 default-router 192.168.21.254
 dns-server 192.168.25.2

- ip dhcp pool VLAN22
 network 192.168.22.0 255.255.255.0
 default-router 192.168.22.254
 dns-server 192.168.25.2

- ip dhcp pool VLAN23
 network 192.168.23.0 255.255.255.0
 default-router 192.168.23.254
```

```
 dns-server 192.168.25.2
- ip dhcp pool VLAN24
 network 192.168.24.0 255.255.255.0
 default-router 192.168.24.254
 dns-server 192.168.25.2

- ip routing

- ipv6 unicast-routing

- interface Port-channel1
 description "range Gigabit Ethernet1/0/2-5"
 switchport trunk allowed vlan 20-25
 switchport mode trunk

- interface GigabitEthernet1/0/1
 description "vers Switch Core 00"
 switchport trunk allowed vlan 20-25
 switchport mode trunk

- interface GigabitEthernet1/0/2
 description "Load-Balancing Protocole LACP"
 switchport trunk allowed vlan 20-25
 switchport mode trunk
 channel-protocol lacp
 channel-group 1 mode active

- interface GigabitEthernet1/0/3
 description "Load-Balancing Protocole LACP"
 switchport trunk allowed vlan 20-25
 switchport mode trunk
 channel-protocol lacp
 channel-group 1 mode active

- interface GigabitEthernet1/0/4
 description "Load-Balancing Protocole LACP"
 switchport trunk allowed vlan 20-25
 switchport mode trunk
 channel-protocol lacp
 channel-group 1 mode active

- interface GigabitEthernet1/0/5
 description "Load-Balancing Protocole LACP"
 switchport trunk allowed vlan 20-25
 switchport mode trunk
```

```
channel-protocol lacp
channel-group 1 mode active
```

- interface GigabitEthernet1/0/6
  - description "vers Switch Serveur Bat Rouge"
  - switchport access vlan 25
  - switchport mode access
- interface GigabitEthernet1/0/7
  - description "vers Switch Informatique"
  - switchport access vlan 20
  - switchport mode access
- interface GigabitEthernet1/0/8
  - description "vers Switch Finance"
  - switchport access vlan 21
  - switchport mode access
- interface GigabitEthernet1/0/9
  - description "vers Switch Commercial"
  - switchport access vlan 22
  - switchport mode access
- interface GigabitEthernet1/0/10
  - description "vers Switch RH"
  - switchport access vlan 23
  - switchport mode access
- interface GigabitEthernet1/0/11
  - description "vers Switch Direction"
  - switchport access vlan 24
  - switchport mode access
- interface Vlan20
  - description "vlan Informatique"
  - mac-address 0002.16b6.4701
  - ip address 192.168.20.252 255.255.255.0
  - ipv6 address 2001:0:CAFE:BE20::252/64
- interface Vlan21
  - description "vlan Finance"
  - mac-address 0002.16b6.4702
  - ip address 192.168.21.252 255.255.255.0
  - ipv6 address 2001:0:CAFE:BE21::252/64



- interface Vlan22
  - description "vlan Commercial"
  - mac-address 0002.16b6.4703
  - ip address 192.168.22.252 255.255.255.0
  - ipv6 address 2001:0:CAFE:BE22::252/64
- interface Vlan23
  - description "vlan RH"
  - mac-address 0002.16b6.4704
  - ip address 192.168.23.252 255.255.255.0
  - ipv6 address 2001:0:CAFE:BE23::252/64
- interface Vlan24
  - description "vlan Direction"
  - mac-address 0002.16b6.4705
  - ip address 192.168.24.252 255.255.255.0
  - ipv6 address 2001:0:CAFE:BE24::252/64
- interface Vlan25
  - description "vlan Serveur Bat Rouge"
  - mac-address 0002.16b6.4706
  - ip address 192.168.25.252 255.255.255.0
  - ipv6 address 2001:0:CAFE:BE25::252/64

## **Switch Distributed 2 :**

- hostname sw-distri-01
- ip dhcp pool VLAN20
  - network 192.168.20.0 255.255.255.0
  - default-router 192.168.20.254
  - dns-server 192.168.25.2
- ip dhcp pool VLAN21
  - network 192.168.21.0 255.255.255.0
  - default-router 192.168.21.254
  - dns-server 192.168.25.2
- ip dhcp pool VLAN22
  - network 192.168.22.0 255.255.255.0
  - default-router 192.168.22.254
  - dns-server 192.168.25.2
- ip dhcp pool VLAN23
  - network 192.168.23.0 255.255.255.0
  - default-router 192.168.23.254
  - dns-server 192.168.25.2
- ip dhcp pool VLAN24
  - network 192.168.24.0 255.255.255.0

```
default-router 192.168.24.254
dns-server 192.168.25.2
```

- ip routing
- ipv6 unicast-routing
- interface Port-channel1

```
description "range GigabitEthernet1/0/2-5"
switchport trunk allowed vlan 20-25
switchport mode trunk
```
- interface GigabitEthernet1/0/1

```
description "vers Switch Core 00"
switchport trunk allowed vlan 20-25
switchport mode trunk
```
- interface GigabitEthernet1/0/2

```
description "Load-balancing Protocole LACP"
switchport trunk allowed vlan 20-25
switchport mode trunk
channel-protocol lacp
channel-group 1 mode active
```
- interface GigabitEthernet1/0/3

```
description "Load-balancing Protocole LACP"
switchport trunk allowed vlan 20-25
switchport mode trunk
channel-protocol lacp
channel-group 1 mode active
```
- interface GigabitEthernet1/0/4

```
description "Load-balancing Protocole LACP"
switchport trunk allowed vlan 20-25
switchport mode trunk
channel-protocol lacp
channel-group 1 mode active
```
- interface GigabitEthernet1/0/5

```
description "Load-balancing Protocole LACP"
switchport trunk allowed vlan 20-25
switchport mode trunk
channel-protocol lacp
channel-group 1 mode active
```

```
- interface GigabitEthernet1/0/6
 description "vers Switch Serveur Bat Rouge"
 switchport access vlan 25
 switchport mode access

- interface GigabitEthernet1/0/7
 description "vers Switch Informatique"
 switchport access vlan 20
 switchport mode access

- interface GigabitEthernet1/0/8
 description "vers Switch Finance"
 switchport access vlan 21
 switchport mode access

- interface GigabitEthernet1/0/9
 description "vers Switch Commercial"
 switchport access vlan 22
 switchport mode access

- interface GigabitEthernet1/0/10
 description "vers Switch RH"
 switchport access vlan 23
 switchport mode access

- interface GigabitEthernet1/0/11
 description "vers Switch Direction"
 switchport access vlan 24
 switchport mode access

- interface Vlan20
 description "vlan Informatique"
 mac-address 00e0.8f82.c601
 ip address 192.168.20.251 255.255.255.0
 ipv6 address 2001:0:CAFE:BE20::251/64

- interface Vlan21
 description "vlan Finance"
 mac-address 00e0.8f82.c602
 ip address 192.168.21.251 255.255.255.0
 ipv6 address 2001:0:CAFE:BE21::251/64

- interface Vlan22
 description "vlan Commercial"
 mac-address 00e0.8f82.c603
```

```
ip address 192.168.22.251 255.255.255.0
ipv6 address 2001:0:CAFE:BE22::251/64
```

```
- interface Vlan23
 description "vlan RH"
 mac-address 00e0.8f82.c604
 ip address 192.168.23.251 255.255.255.0
 ipv6 address 2001:0:CAFE:BE23::251/64
```

```
- interface Vlan24
 description "vlan Direction"
 mac-address 00e0.8f82.c605
 ip address 192.168.24.251 255.255.255.0
 ipv6 address 2001:0:CAFE:BE24::251/64
```

```
- interface Vlan25
 description "vlan Serveur Bat Rouge"
 mac-address 00e0.8f82.c606
 ip address 192.168.25.251 255.255.255.0
 ipv6 address 2001:0:CAFE:BE25::251/64
```

#### **4.2 Configuration des switches de type layer 2 :**

##### **Switch Serveur WEB :**

```
- hostname sw-serveur-bat-vert
- interface FastEthernet0/1
 description "vers Serveur WEB"
 switchport mode access
- interface GigabitEthernet0/1
 description "vers Routeur Bat Vert"
 switchport mode access
```

##### **Switch Réunion:**

```
- hostname sw-reunion
- interface FastEthernet0/1
 description "vers Reunion"
 switchport mode access
- interface GigabitEthernet0/1
 description "vers Routeur Bat Vert"
 switchport mode access
```

**Switch Accueil:**

- hostname sw-accueil
- interface FastEthernet0/1
  - description "vers Accueil"
  - switchport mode access
- interface GigabitEthernet0/1
  - description "vers Routeur Bat Bleu"
  - switchport mode access

**Switch Showroom:**

- hostname sw-showroom
- interface FastEthernet0/1
  - description "vers Showroom"
  - switchport mode access
- interface GigabitEthernet0/1
  - description "vers Routeur Bat Bleu"
  - switchport mode access

**Switch Direction:**

- hostname sw-direction
- interface FastEthernet0/1
  - description "vers Direction"
  - switchport access vlan 24
  - switchport mode access
- interface GigabitEthernet0/1
  - description "vers Switch Distributed 00"
  - switchport access vlan 24
  - switchport mode access
- interface GigabitEthernet0/2
  - description "vers Switch Distributed 01"
  - switchport access vlan 24
  - switchport mode access

**Switch RH:**

- hostname sw-rh
- interface FastEthernet0/1

```
description "vers RH"
switchport access vlan 23
switchport mode access
```

```
- interface GigabitEthernet0/1
 description "vers Switch Distributed 00"
 switchport access vlan 23
 switchport mode access
```

```
- interface GigabitEthernet0/2
 description "vers Switch Distributed 01"
 switchport access vlan 23
 switchport mode access
```

#### **Switch Commerciaux/Chargés de clientèle:**

```
- hostname sw-commercial
```

```
- interface FastEthernet0/1
 description "vers Commercial"
 switchport access vlan 22
 switchport mode access
```

```
- interface GigabitEthernet0/1
 description "vers Switch Distributed 00"
 switchport access vlan 22
 switchport mode access
```

```
- interface GigabitEthernet0/2
 description "vers Switch Distributed 01"
 switchport access vlan 22
 switchport mode access
```

#### **Switch Finance :**

```
- hostname sw-finance
```

```
- interface FastEthernet0/1
 description "vers Finance"
 switchport access vlan 21
 switchport mode access
```

```
- interface GigabitEthernet0/1
 description "vers Switch Distributed 00"
 switchport access vlan 21
 switchport mode access
```

- interface GigabitEthernet0/2
  - description "vers Switch Distributed 01"
  - switchport access vlan 21
  - switchport mode access

#### **Switch Service Informatique :**

- hostname sw-informatique
- interface FastEthernet0/1
  - description "vers Informatique"
  - switchport access vlan 20
  - switchport mode access
- interface GigabitEthernet0/1
  - description "vers Switch Distributed 00"
  - switchport access vlan 20
  - switchport mode access
- interface GigabitEthernet0/2
  - description "vers Switch Distributed 01"
  - switchport access vlan 20
  - switchport mode access

#### **Switch du serveur DNS :**

hostname sw-serveur-bat-rouge

- interface FastEthernet0/1
  - description "vers Serveur DNS"
  - switchport access vlan 25
  - switchport mode access
- interface GigabitEthernet0/1
  - description "vers Switch Distributed 00"
  - switchport access vlan 25
  - switchport mode access
- interface GigabitEthernet0/2
  - description "vers Switch Distributed 01"
  - switchport access vlan 25
  - switchport mode access